



1. Description

1.1. Project

Project Name	mtk3bsp2_stm32n657
Board Name	custom
Generated with:	STM32CubeMX 6.18.0
Date	08/07/2026

1.2. MCU

MCU Series	STM32N6
MCU Line	STM32N6x7
MCU name	STM32N657X0HxQ
MCU Package	VFBGA264
MCU Pin number	264

1.3. Core(s) information

Core(s)	ARM Cortex-M55
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3. Pins Configuration

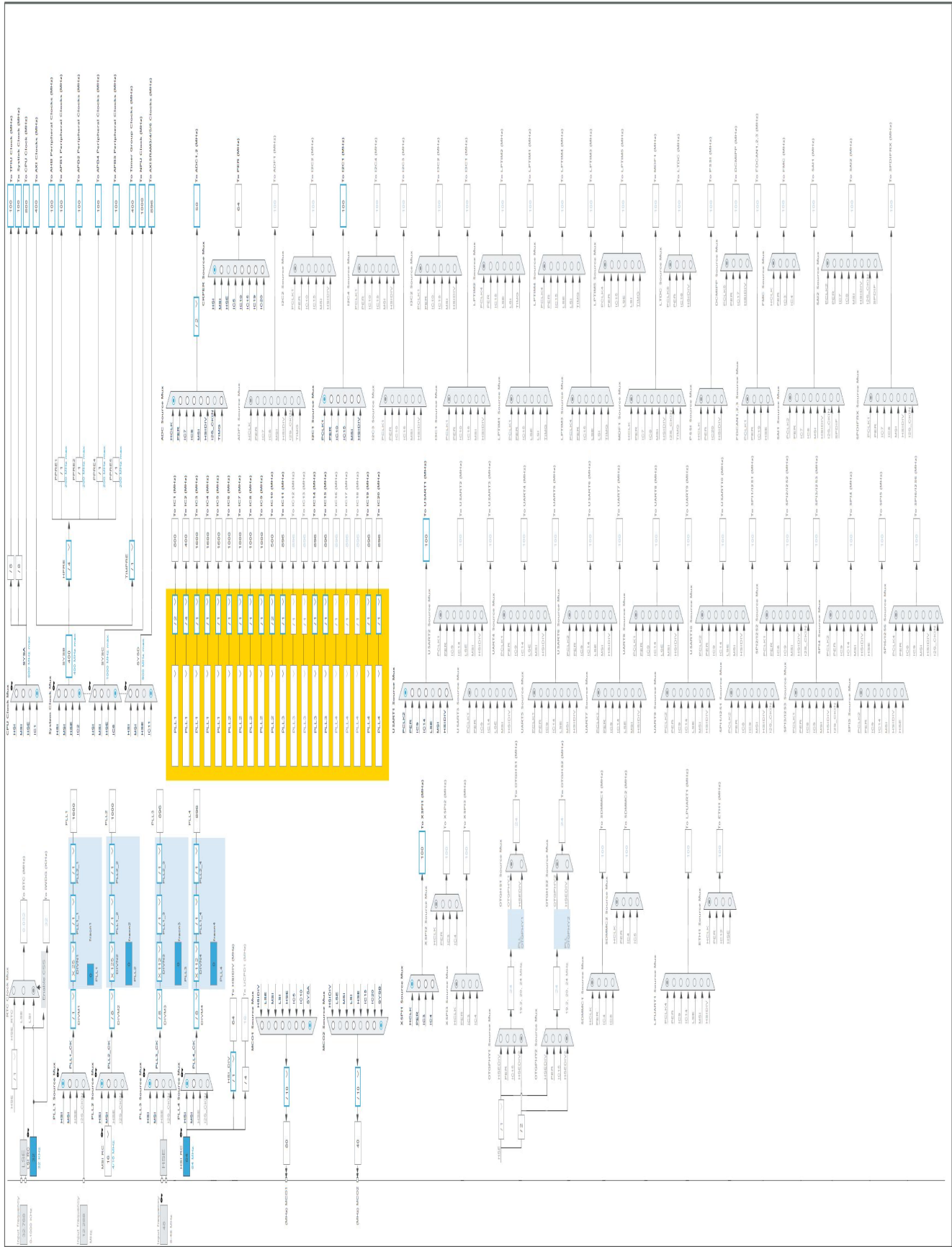
Pin Number VFBGA264	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	PDR_ON	MonoIO		
A4	VDD33USB	Power		
A9	PC1	I/O	I2C1_SDA	
A10	PC4	I/O	USART1_TX	
A17	VSS	Power		
B4	NC	NC		
C4	NC	NC		
C5	VDDA18USB	Power		
C6	OTG1_TXRTUNE	MonoIO		
C7	PC9	I/O	RCC_MCO_2	
C11	PE5	I/O	I2C1_SCL	
D6	OTG2_TXRTUNE	MonoIO		
D11	PE6	I/O	USART1_RX	
E1	VBAT	Power		
E2	PWR_ON	MonoIO		
E8	VDDIO4	Power		
E9	VDDIO5	Power		
F1	V08CAP	Power		
F2	VSSAPMU	Power		
F6	VDDA18AON	Power		
F7	VSS	Power		
F8	VDDIO4	Power		
F9	VDDIO5	Power		
F10	VSS	Power		
F11	VSS	Power		
F12	VSS	Power		
G1	VDDA18PMU	Power		
G2	VFBSMPS	Power		
G3	NRST	Reset		
G4	BOOT0	Boot		
G6	VSSAON	Power		
G12	VSS	Power		
G13	VDD	Power		
G15	PP7	I/O	XSPIM_P1_IO7	
G16	PP6	I/O	XSPIM_P1_IO6	
G17	PP0	I/O	XSPIM_P1_IO0	

Pin Number VFBGA264	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
H1	VSSSMPS	Power		
H2	VSSSMPS	Power		
H3	VSSSMPS	Power		
H4	VSSSMPS	Power		
H5	VSSSMPS	Power		
H6	VSSSMPS	Power		
H12	VDD	Power		
H13	VDD	Power		
H15	PP4	I/O	XSPIM_P1_IO4	
H16	PP1	I/O	XSPIM_P1_IO1	
J1	VLXSMPS	Power		
J2	VLXSMPS	Power		
J3	VLXSMPS	Power		
J4	VLXSMPS	Power		
J5	VLXSMPS	Power		
J6	VLXSMPS	Power		
J12	VDD	Power		
J13	VDD	Power		
J14	PP5	I/O	XSPIM_P1_IO5	
J16	PP3	I/O	XSPIM_P1_IO3	
J17	PP2	I/O	XSPIM_P1_IO2	
K1	VDDSMPS	Power		
K2	VDDSMPS	Power		
K3	VDDSMPS	Power		
K4	VDDSMPS	Power		
K5	VDDSMPS	Power		
K6	VDDSMPS	Power		
K12	VSS	Power		
K13	VDD	Power		
K17	PO2	I/O	XSPIM_P1_DQS0	
L2	PF4 *	I/O	GPIO_Output	SMPS_OVD
L6	VDDCORE	Power		
L12	VDDCORE	Power		
L13	VDDIO2	Power		
L14	VDDIO2	Power		
M5	VDDA18PLL	Power		
M6	VSSA	Power		
M7	VDDCORE	Power		
M8	VSS	Power		

Pin Number VFBGA264	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
M9	VDDCORE	Power		
M10	VSS	Power		
M11	VDDCORE	Power		
M12	VSS	Power		
M13	VSS	Power		
M14	VDDIO2	Power		
M15	VDDIO2	Power		
N5	VSS	Power		
N6	VDDA18ADC	Power		
N7	VSS	Power		
N8	VDDCORE	Power		
N9	VSS	Power		
N10	VDDCORE	Power		
N11	VSS	Power		
N12	VDDCORE	Power		
N13	VDDIO3	Power		
N16	PO4	I/O	XSPIM_P1_CLK	
P7	PA9	I/O	ADC1_INP10	
P11	VDDCORE	Power		
P13	VDDIO3	Power		
R4	VDDCSI	Power		
R5	VDDA18CSI	Power		
R8	PA5	I/O	ADC2_INP18	
T2	VREF-	Power		
U1	VSS	Power		
U2	VREF+	MonoIO		
U8	PA8	I/O	RCC_MCO_1	
U17	VSS	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32N6
Line	STM32N6x7
MCU	STM32N657X0HxQ
Datasheet	DS00002 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	1.8

1.3. Battery Selection

Battery	Li-SOCL2(DD36000)
Capacity	36000.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	450.0 mA
Max Pulse Current	1000.0 mA
Cells in series	1
Cells in parallel	1

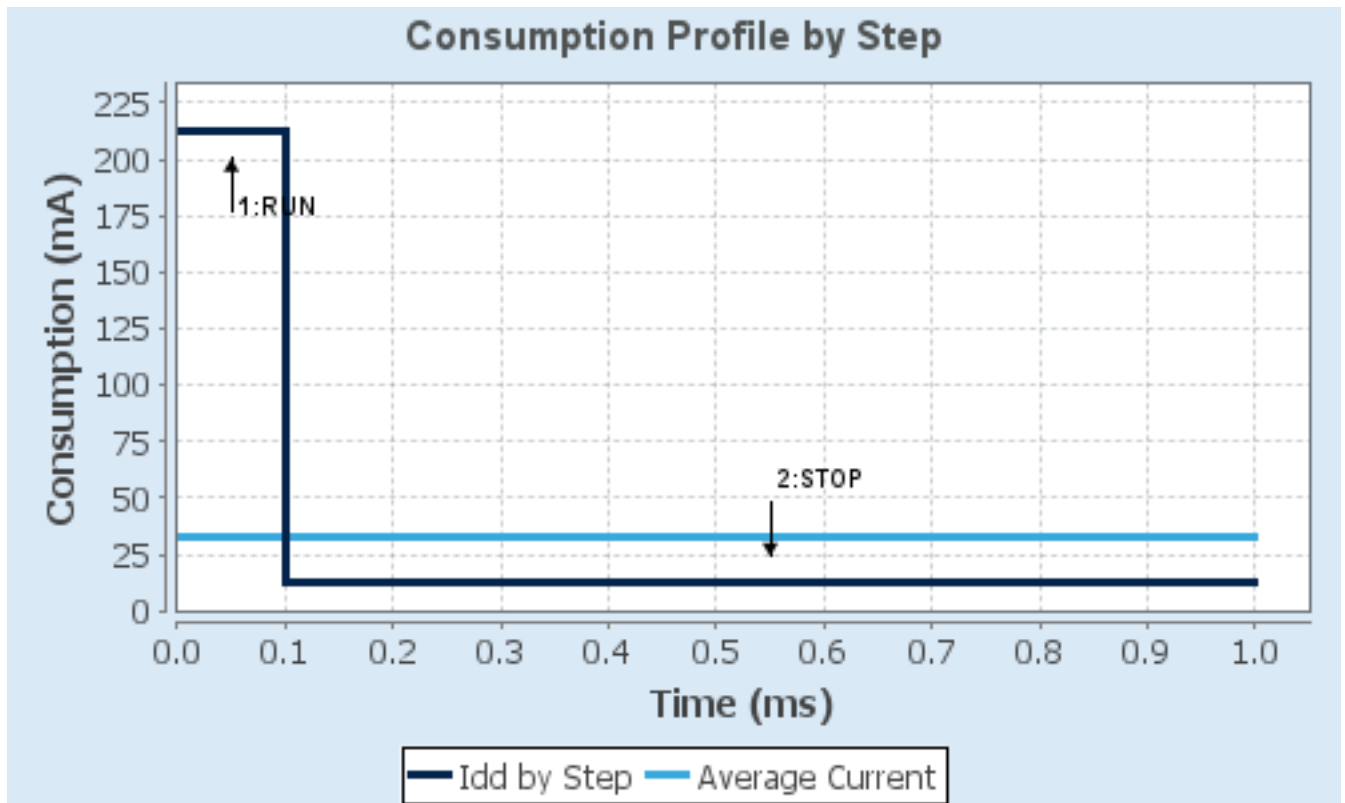
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	1.8	1.8
Voltage Source	Battery	Battery
Range	VOS_HIGH/extern SMPS	SVOS_LOW/intern SMPS
Fetch Type	AXI-SRAM2/CACHE-ON	FLASH
CPU Frequency	800 Hz	0 Hz
Clock Configuration	HSI PLL:400/800 ALL_IPs_ON	ALL CLOCKS OFF
Clock Source Frequency	400 Hz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	212.16 mA	12.3 mA
Duration	0.1 ms	0.9 ms
DMIPS	1.0	0.0
Ta Max	90.25	122.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	32.29 mA
Battery Life	1 month, 15 days, 23 hours	Average DMIPS	0.10239999 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	mtk3bsp2_stm32n657
Project Folder	D:\mtk3bsp2_stm32n657\mtk3bsp2_stm32n657
Toolchain / IDE	EWARM V9.40
Firmware Package Name and Version	STM32Cube FW_N6 V1.4.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	FSBL-0x200
Minimum Stack Size	FSBL-0x800

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls FSBL

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ICACHE_Init	ICACHE
4	MX_XSPI1_Init	XSPI1
5	MX_XSPIM_Init	XSPIM
6	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

2.4. Advanced Settings - Generated Function Calls Appli

Rank	Function Name	Peripheral Instance Name
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Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_USART1_UART_Init	USART1
3	MX_ADC1_Init	ADC1
4	MX_ADC2_Init	ADC2
5	MX_XSPI1_Init	XSPI1
6	MX_XSPIM_Init	XSPIM
7	MX_I2C1_Init	I2C1
8	SystemIsolation_Config	RIF

2.5. Advanced Settings - Generated Function Calls ExtMemLoader

Rank	Function Name	Peripheral Instance Name
1	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN10: IN10 Single-ended

3.1.1. Parameter Settings:

Core(s) Settings:

Context(s):

Application

ADCs_Common_Settings:

Mode

Independent mode

ADC_Settings:

Resolution

ADC 12-bit resolution

Scan Conversion Mode

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

Gain Compensation

0

End Of Conversion Selection

End of single conversion

Overrun behaviour

Overrun data preserved

Conversion Data Management Mode

Regular Conversion data stored in DR register only

Low Power Auto Wait

Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Left Bit Shift

No bit shift

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Regular Conversion launched by software

External Trigger Conversion Edge

None

Rank

1

Channel

Channel 10

Sampling Time

1.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

3.2. ADC2

mode: IN18

3.2.1. Parameter Settings:

Core(s) Settings:

Context(s):

Application

ADCs_Common_Settings:

Mode

Independent mode

ADC_Settings:

Resolution

ADC 12-bit resolution

Scan Conversion Mode

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

Gain Compensation

0

End Of Conversion Selection

End of single conversion

Overrun behaviour

Overrun data preserved

Conversion Data Management Mode

Regular Conversion data stored in DR register only

Low Power Auto Wait

Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Left Bit Shift

No bit shift

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Regular Conversion launched by software

External Trigger Conversion Edge

None

Rank

1

Channel

Channel 18

Sampling Time

1.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

3.3. CORTEX_M55_FSBL

3.3.1. Parameter Settings:

Core(s) Settings:

Context(s): First Stage Boot Loader

Cortex Interface Settings:

CPU ICache Disabled

CPU DCache Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode MPU NOT USED

3.3.2. Attributes Settings:

Core(s) Settings:

Context(s): First Stage Boot Loader

Attributes Settings:

3.4. I2C1

I2C: I2C

3.4.1. Parameter Settings:

Core(s) Settings:

Context(s): Application

Timing configuration:

I2C Speed Mode Standard Mode

I2C Speed Frequency (KHz) 100

Rise Time (ns) 0

Fall Time (ns) 0

Coefficient of Digital Filter 0

Analog Filter Enabled

Timing 0x10C0ECFF *

Slave Features:

Clock No Stretch Mode Disabled

General Call Address Detection Disabled

Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.5. ICACHE

Mode: 1-way (direct mapped cache)

Secure Attribute: not secured

3.5.1. Core(s) Settings:

Context(s):	First Stage Boot Loader
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3.6. RCC

mode: Master Clock Output 1

mode: Master Clock Output 2

3.6.1. Parameter Settings:

Core(s) Settings:

Context(s):	First Stage Boot Loader
	Application
	External Memory Loader

Power Parameters:

SupplySource	PWR_EXTERNAL_SOURCE_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

Spread spectrum mode:

PLL1 CSG mode	Disabled
PLL2 CSG mode	Disabled
PLL3 CSG mode	Disabled
PLL4 CSG mode	Disabled

RCC Parameters:

HSI Calibration Value	16
MSI Calibration Value	16
TIM Group1 Prescaler Selection	Disabled
TIM Group2 Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

System Parameters:

VDD voltage (V) 3.3

3.7. RIF

mode: RISUP

3.7.1. Parameter Settings:

Core(s) Settings:

Context(s):

Application

0

SPI1/I2S1

false

false

-

false

1

SPI2/I2S2

false

false

-

false

2

SPI3/I2S3

false

false

-

false

3

SPI4

false

false

-

false

4

SPI5

false

false

-

false

5

SPI6/I2S6

false

false

-

false

6

SAI1

false

	false
	-
	false
8	
SAI2	false
	false
	-
	false
9	
I2C1	false
	false
	-
	false
10	
I2C2	false
	false
	-
	false
11	
I2C3	false
	false
	-
	false
12	
I2C4	false
	false
	-
	false
13	
I3C1	false
	false
	-
	false
14	
I3C2	false
	false
	-
	false
15	
USART1	false
	false
	-
	false

16		
USART2	false	
	false	
	-	
	false	
17		
USART3	false	
	false	
	-	
	false	
18		
UART4	false	
	false	
	-	
	false	
19		
UART5	false	
	false	
	-	
	false	
20		
USART6	false	
	false	
	-	
	false	
21		
UART7	false	
	false	
	-	
	false	
22		
UART8	false	
	false	
	-	
	false	
23		
UART9	false	
	false	
	-	
	false	
24		
USART10	false	
	false	

	-
	false
25	
LPUART1	false
	false
	-
	false
26	
FDCAN1	false
	false
	-
	false
26	
FDCAN2	false
	false
	-
	false
26	
FDCAN3	false
	false
	-
	false
27	
TIM1	false
	false
	-
	false
28	
TIM2	false
	false
	-
	false
29	
TIM3	false
	false
	-
	false
30	
TIM4	false
	false
	-
	false
31	

TIM5	false
	false
	-
	false
32	
TIM6	false
	false
	-
	false
33	
TIM7	false
	false
	-
	false
34	
TIM8	false
	false
	-
	false
35	
TIM9	false
	false
	-
	false
36	
TIM10	false
	false
	-
	false
37	
TIM11	false
	false
	-
	false
38	
TIM12	false
	false
	-
	false
39	
TIM13	false
	false
	-

	false
40	
TIM14	false
	false
	-
	false
41	
TIM15	false
	false
	-
	false
42	
TIM16	false
	false
	-
	false
43	
TIM17	false
	false
	-
	false
44	
TIM18	false
	false
	-
	false
45	
GFXTIM	false
	false
	-
	false
46	
LPTIM1	false
	false
	-
	false
47	
LPTIM2	false
	false
	-
	false
48	
LPTIM3	false

	false
	-
	false
49	
LPTIM4	false
	false
	-
	false
50	
LPTIM5	false
	false
	-
	false
51	
ADF1	false
	false
	-
	false
52	
MDF1	false
	false
	-
	false
53	
SDMMC1	false
	false
	-
	false
54	
SDMMC2	false
	false
	-
	false
55	
MDIOS	false
	false
	-
	false
56	
OTG1_HS	false
	false
	-
	false

57		
OTG2_HS	false	
	false	
	-	
	false	
58		
UCPD1	false	
	false	
	-	
	false	
60		
ETH1	false	
	false	
	-	
	false	
61		
SPDIFRX1	false	
	false	
	-	
	false	
62		
SYSCFG	false	
	false	
	-	
	false	
64		
ADC12	false	
	false	
	-	
	false	
65		
VREFBUF	false	
	false	
	-	
	false	
67		
CRC	false	
	false	
	-	
	false	
68		
IWDG	false	
	false	

	-
	false
69	
WWDG	false
	false
	-
	false
76	
RNG	false
	false
	-
	false
77	
PKA	false
	false
	-
	false
78	
SAES	false
	false
	-
	false
79	
HASH	false
	false
	-
	false
80	
CRYP	false
	false
	-
	false
81	
MCE1	false
	false
	-
	false
82	
MCE2	false
	false
	-
	false
83	

MCE3	false
	false
	-
	false
84	
MCE4	false
	false
	-
	false
86	
XSPI1	false
	false
	-
	false
87	
XSPI2	false
	false
	-
	false
88	
XSPI3	false
	false
	-
	false
89	
XSPIM	false
	false
	-
	false
90	
FMC	false
	false
	-
	false
92	
CSI2HOST	false
	false
	-
	false
93	
DCMIPP	false
	false
	-

	false
94	
DCMI	false
	false
	-
	false
96	
JPEG	false
	false
	-
	false
97	
VENC	false
	false
	-
	false
98	
ICACHE	false
	false
	-
	false
99	
GPU	false
	false
	-
	false
100	
GFXMMU	false
	false
	-
	false
101	
DMA2D	false
	false
	-
	false
102	
LTDC_CMN	false
	false
	-
	false
103	
LTDC_L1	false

	false
	-
	false
104	
LTDC_L2	false
	false
	-
	false
106	
NPU	false
	false
	-
	false

3.8. USART1

Mode: Asynchronous

3.8.1. Parameter Settings:

Core(s) Settings:

Context(s):

Application

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration
Autonomous Mode	Disable

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable

Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.9. VREFBUF

VREFBUF Mode: Internal voltage reference

3.9.1. Parameter Settings:

Core(s) Settings:

Context(s):	Application
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Voltage_Reference_Buffer_Settings:

Trimming Mode	Factory Trimming
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3.10. XSPI1

Mode: Octo SPI

Port: Port1 Octo

3.10.1. Parameter Settings:

Core(s) Settings:

Context(s):	First Stage Boot Loader Application
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Generic:

Fifo Threshold	1
Memory Mode	Disable
Memory Type	Micron
Memory Size	16 Bits
Chip Select High Time Cycle	1
Free Running Clock	Disable
Clock Mode	Low
Wrap Size	Not Supported
Clock Prescaler	0
Sample Shifting	None
Delay Hold Quarter Cycle	Disable
Chip Select Boundary	Disabled
Maximum Transfer	0
Refresh Rate	0
Memory Select	NCS1

Switching Duration Clock Number

1

3.11. XSPIM

Mode: Direct (XSPI1 to Port1; XSPI2 to Port2; XSPI3 not used)

3.11.1. Core(s) Settings:

Context(s):

First Stage Boot Loader
Application

3.12. EXTMEM_MANAGER

mode: Activate External Memory Manager

3.12.1. Boot usecase:

Core(s) Settings:

Context(s):

First Stage Boot Loader
External Memory Loader

Boot:

Select boot code generation

true *

Selection of the boot system

Execute In Place

Header size

0x400 *

XIP:

select the memory

Memory 1

Application offset

0x100000 *

3.12.2. Memory 1:

Core(s) Settings:

Context(s):

First Stage Boot Loader
External Memory Loader

Select driver:

Select the memory driver

EXTMEM_NOR_SFDP

Configuration:

Memory Instance

XSPI1

Number of memory data lines

EXTMEM_LINK_CONFIG_1LINE

3.12.3. Memory 2:

Core(s) Settings:

Context(s):

First Stage Boot Loader
External Memory Loader

Select driver:

Select the memory driver

NONE

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
ADC1	PA9	ADC1_INP10	Analog mode	No pull-up and no pull-down	n/a		Application
ADC2	PA5	ADC2_INP18	Analog mode	No pull-up and no pull-down	n/a		Application
I2C1	PC1	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Application
	PE5	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Application
RCC	PC9	RCC_MCO_2	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		First Stage Boot Loader Application
	PA8	RCC_MCO_1	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		First Stage Boot Loader Application
USART1	PC4	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Application
	PE6	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Application
XSPI1	PP7	XSPIM_P1_IO 7	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PP6	XSPIM_P1_IO 6	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PP0	XSPIM_P1_IO 0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PP4	XSPIM_P1_IO 4	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PP1	XSPIM_P1_IO 1	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PP5	XSPIM_P1_IO 5	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PP3	XSPIM_P1_IO 3	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PP2	XSPIM_P1_IO 2	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
	PO2	XSPIM_P1_D	Alternate Function Push	No pull-up and no pull-down	Very High		First Stage Boot

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
		QS0	Pull				Loader Application
	PO4	XSPIM_P1_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High		First Stage Boot Loader Application
GPIO	PF4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SMPS_OVD	First Stage Boot Loader* Application

* Initialized context

4.2. GPDMA1

4.2.1. Core(s) Settings:

Context(s): First Stage Boot Loader Application

4.3. HPDMA1

4.3.1. Core(s) Settings:

Context(s): First Stage Boot Loader Application

4.4. LINKEDLIST

4.4.1. Core(s) Settings:

Context(s): First Stage Boot Loader Application

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
Secure fault	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
FPU global interrupt	unused		
RCC secure global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
I2C1 Event interrupt	unused		
I2C1 Error interrupt	unused		
USART1 global interrupt	unused		
XSPI1 global interrupt	unused		

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
Secure fault	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0

Interrupt Table	Enable	Preenmption Priority	SubPriority
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
Secure fault	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
FPU global interrupt	unused		
RCC secure global interrupt	unused		
GPU2D cache interrupt	unused		
XSPI1 global interrupt	unused		

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
Secure fault	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Category view Context Execution view

Choose filters ... by Context Execution

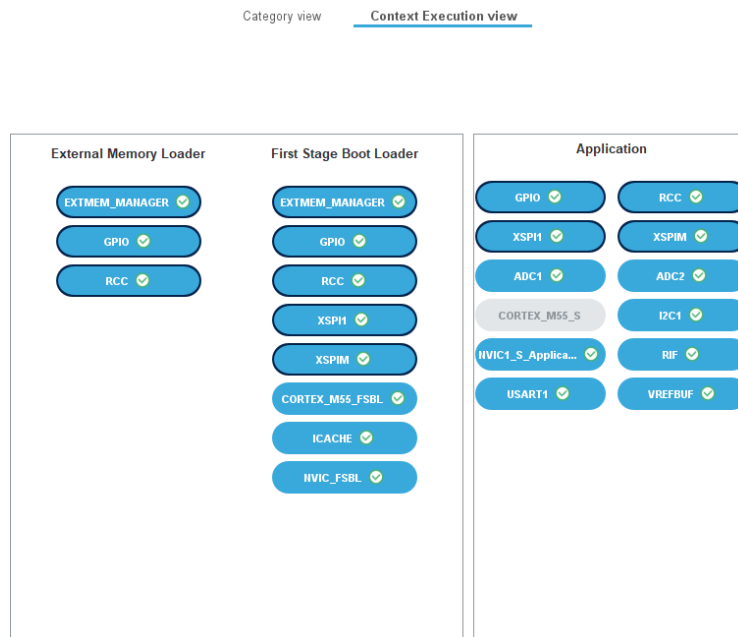
☐ First Stage Boot Loader ☐ Application ☐ External Memory Loader

Middleware

EXTMEM_MAHAGER ✓

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Other
CORTEX_M55_FSBL ✓	ADC1 ✓		I2C1 ✓		RIF ✓				LINKEDLIST	
CORTEX_M55_S	ADC2 ✓		USART1 ✓							
GPDMA1	VREFBUF ✓		XSPH ✓							
GPIO ✓			XSPIM ✓							
HPDMA1										
ICACHE ✓										
NVIC1_S_Applicati... ✓										
NVIC_FSBL ✓										
RCC ✓										

5.2. Context Execution view



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32n6-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32n6-ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32n6-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32cubeai_press_pres.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32n6-series-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/st-neural-art-accelerator-introduction.pdf
Presentations	https://www.st.com/resource/en/product_presentation/presentation-st-edge-ai-suite.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32n6.pdf
Security Advisory	https://www.st.com/resource/en/security_advisory/sa0054-stm32cube-software-mce-driver-encryption-with-zero-value-key-in-stm32n6-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-

	stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0067-applicability-of-threadx-middleware-vulnerabilities-on-stm32cube-packages-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdf-tft-display-controller-lt-dc-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5593-how-to-use-the-gpdma-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5879-introduction-to-i3c-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5020-introduction-to

digital-camera-interface-dcml-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5967-getting-started-with-hardware-development-for-stm32n6-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5946-how-to-optimize-lowpower-modes-on-stm32n6-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6000-how-to-build-the-discrete-power-supply-for-stm32n6-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6197-stm32cube-mcu-package-examples-for-stm32n6-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6211-introduction-to-digital-camera-interface-pixel-pipeline-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6228-how-to-implement-xspi-readwhilewrite-feature-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-how-to-use-dac-for-audio-and-waveform-generation-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-how-to-use-generalpurpose-timer-peripheral-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4989-introduction-to-debug-toolbox-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5050-getting-started-

	with-octospi-hexadecapi-and-xspi-interfaces-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5795-how-to-use-multifunction-digital-filter-mdf-to-capture-sound-on-stm32-mcus-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0620-stm32n6xxx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm01125716.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0273-stm32-cortexm55-mcus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0486-stm32n647657xx-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-devices-in-bga-packages-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3234-how-to-proceed-

with-boot-rom-on-stm32n6-mcus-stmicroelectronics.pdf

User Manuals https://www.st.com/resource/en/user_manual/um3249-getting-started-with-stm32cuben6-for-stm32n6-series-stmicroelectronics.pdf

User Manuals https://www.st.com/resource/en/user_manual/um3451-stm32n6xx-security-guidance-for-sesip-level-3-certification-stmicroelectronics.pdf